

GE Healthcare

Signa HDx Forward Production B–Port Connector Upgrade

Technical Publication



5241502
Rev 1

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “**damage in shipment**” written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1–800–548–3366 and use option 8.
- Contact your local service coordinator for more information on this process.

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WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

- ・このサービスマニュアルには英語版しかありません。
- ・GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- ・このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
- ・この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意：

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

Revision History

Date	Rev	Change Description
6-6-2007	1	Initial Release

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SECTION 1 – OVERVIEW

1–1 INTRODUCTION

This manual provides instructions to upgrade a forward production Signa HDx system with a B–Port connector in the LPCA.

This Direction covers the following catalog:

M3335AY – B–Port Connector Upgrade Kit

This upgrade adds:

- B8 Bezel
- Cable Track with associated cables
- One cable between the 16 Channel Switch and the Image Reject Filter

1–2 SITE READY CHECK

Pre–installation work must be completed before equipment is delivered to avoid delays and confusion.

1–3 DISCARDED MATERIAL RETURN POLICY

The equipment or material being removed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

1–4 PRODUCT LOCATOR

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans–shipment, and field location of the serialized models. There are now two methods for submitting Product Locator information.

1. At this time, for **U.S. ONLY**, the preferred method of submitting information is the *FE Site Verification Web Site* at:
<http://gein2.med.ge.com/gib>

The FE Site Verification consists of three components that are available on the web from the main menu. They are:

- Install/deinstall product locator model and serial numbers
- Add/modify ship to address information
- Update CARES FE data for primary/secondary FEs

To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following:
Product Locator Support Central Page – <http://supportcentral.ge.com/15563>

2. One “Shipping Card” is filled out and submitted when shipped (extra cards are supplied for trans–shipments between storage and distribution points), and the “Installation Card” and extra shipment cards are attached.

1-4 PRODUCT LOCATOR (Continued)

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

1-5 FINAL COORDINATION

1. Replace all covers that have not been previously replaced.
2. Record and enter applicable data into applicable site configuration files and records.
3. Process product locator installation cards for all serialized components added to the system to include product locator card supplied with the upgrade serial number rating plate. Return to:

**Product Locator File
P.O. Box 414, W-523
Milwaukee, WI 53201-0414**

Note

Failure to fill out and return Product Locator Cards may result in failure of your site to receive future FMI's.

SECTION 2 – UPGRADE PROCEDURES

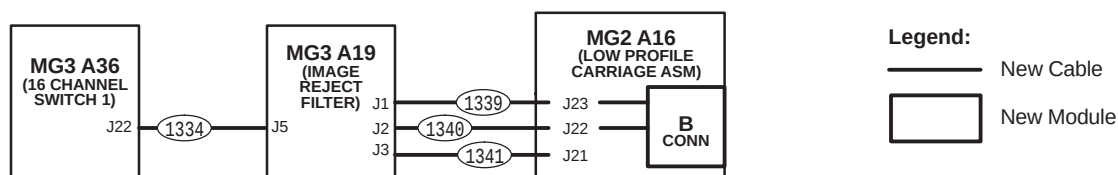
This section contains upgrade procedures for installing the 8-Channel B-Port connector, cable track, and related cables.

2-1 CABLE RUN INFORMATION

The table below provides information for connecting the cables provided in M3335AY.

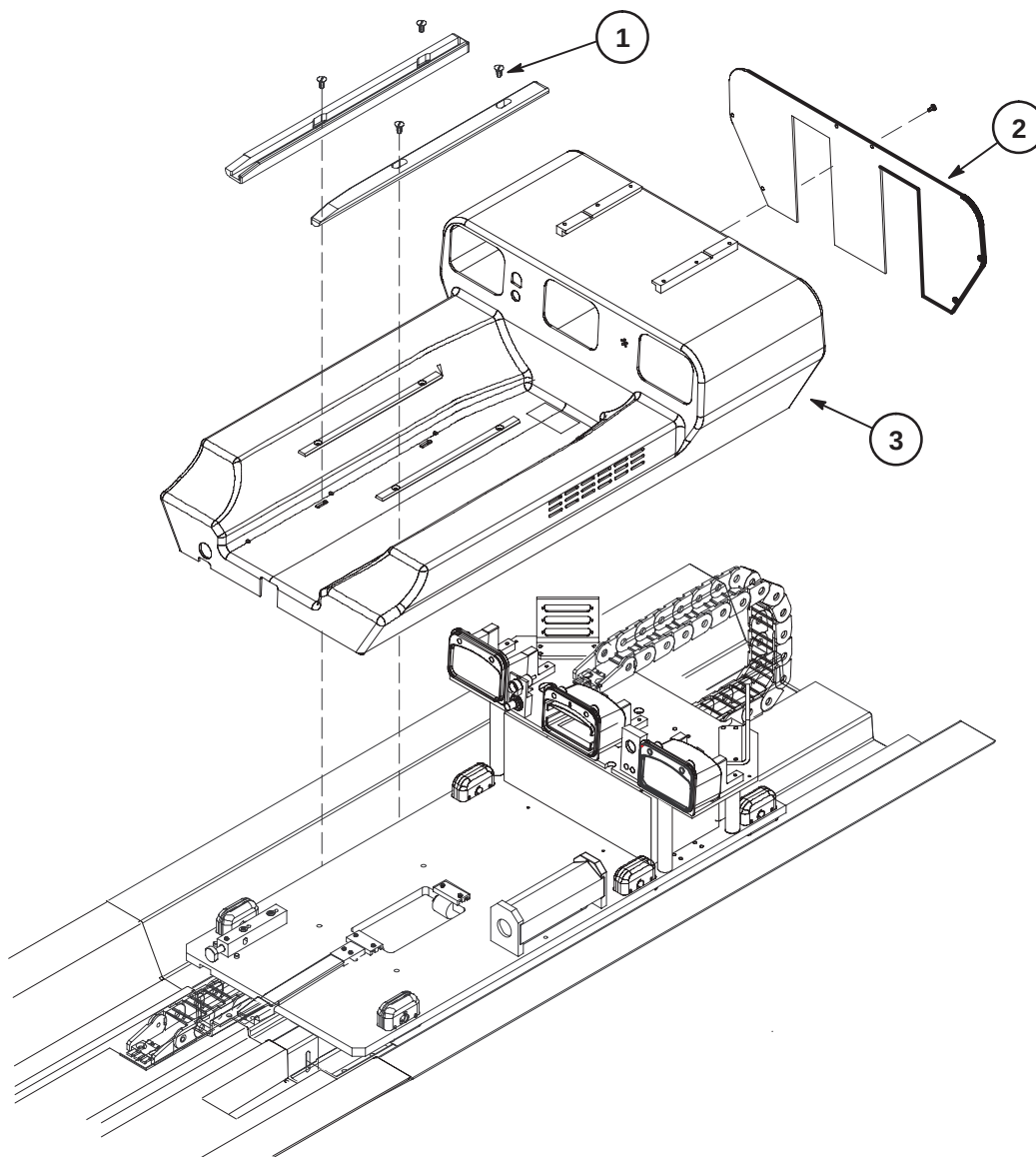
Run Number	Part Number	Connection A	Connection B
1334	5141244	Image Reject #1 J5	16 Ch Sw #1 J22
1339	5141980	Image Reject #1 J1	LPCA-J23
1340	5141979	Image Reject #1 J2	LPCA-J22
1341	5141980-2	Image Reject #1 J3	LPCA-J21 (not used)

Upgrade Cable Interconnect Map



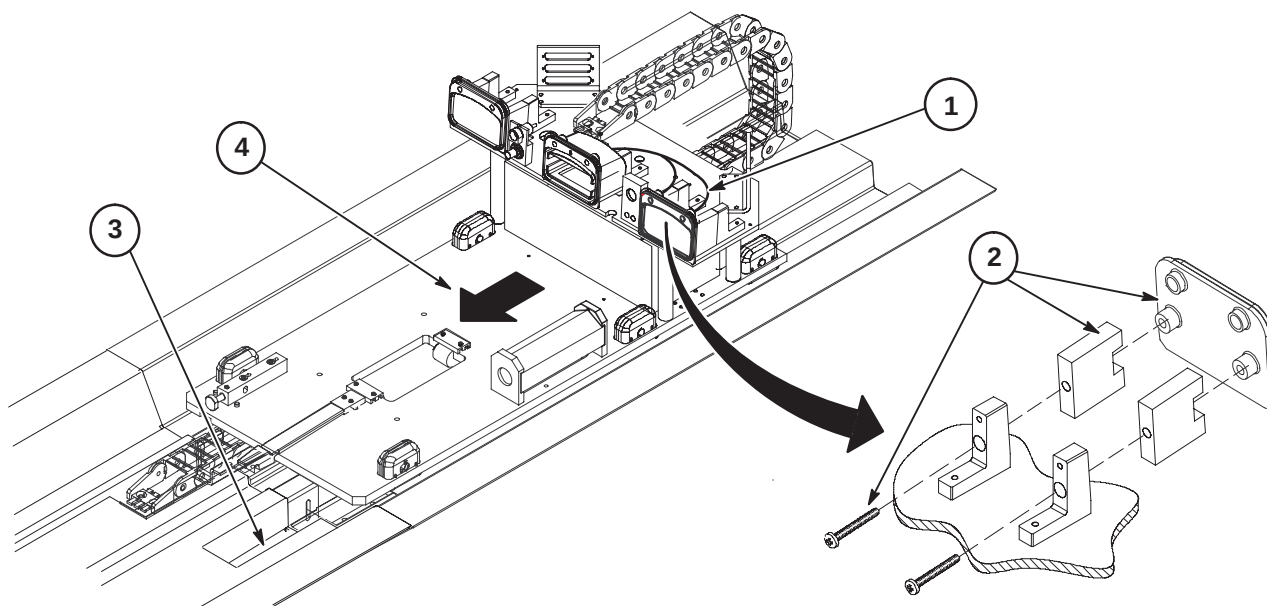
2-2 REMOVE LPCA COVER

1. Unfasten the four screws that attach the cover to the base plate.
2. Unfasten 6 screws and remove Rear Cover from LPCA.
3. Carefully remove cover from base plate assembly.



2-3 DE-INSTALL THE B-PART BLANKING PLATE AND THE CABLE TRACK COVER PLATE

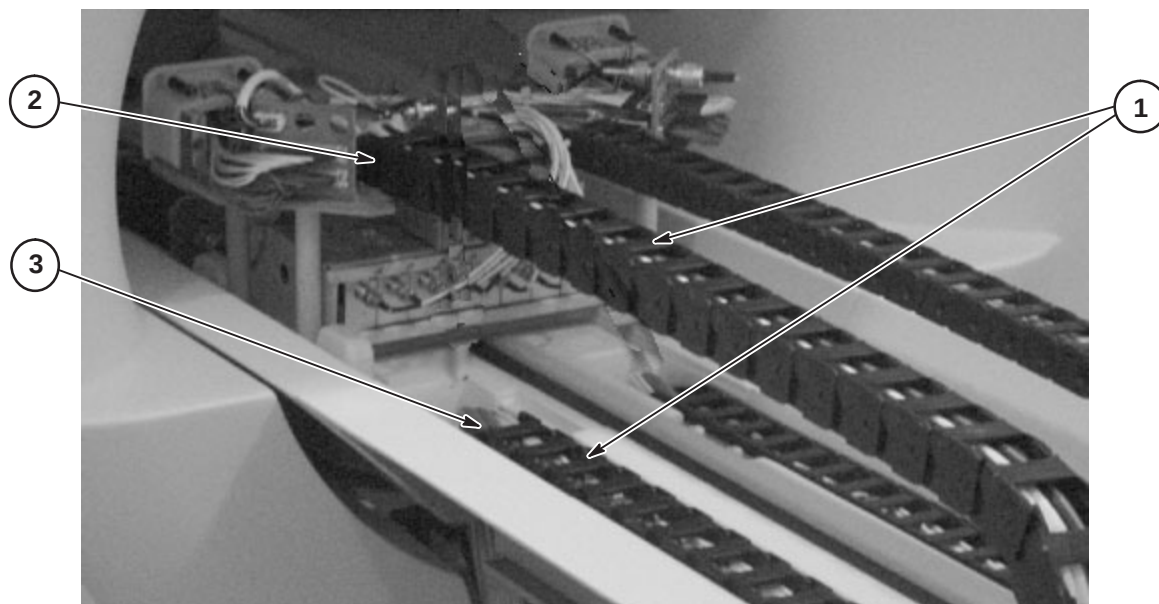
1. Disconnect and retain LED cables from back of Bezel Cover.
2. Unfasten and discard the existing Bezel Cover, Spacer, and M5 x 20mm screws.
3. Remove and discard Cable Track Opening Cover. Retain screws for installation of new Cable Track.
4. Move LPCA forward into bore so rear of LPCA plate is located just past the cable track opening in the rear of the Front Split Bridge.



2-4 INSTALL CABLE TRACK AND RUNS 1339, 1340, AND 1341

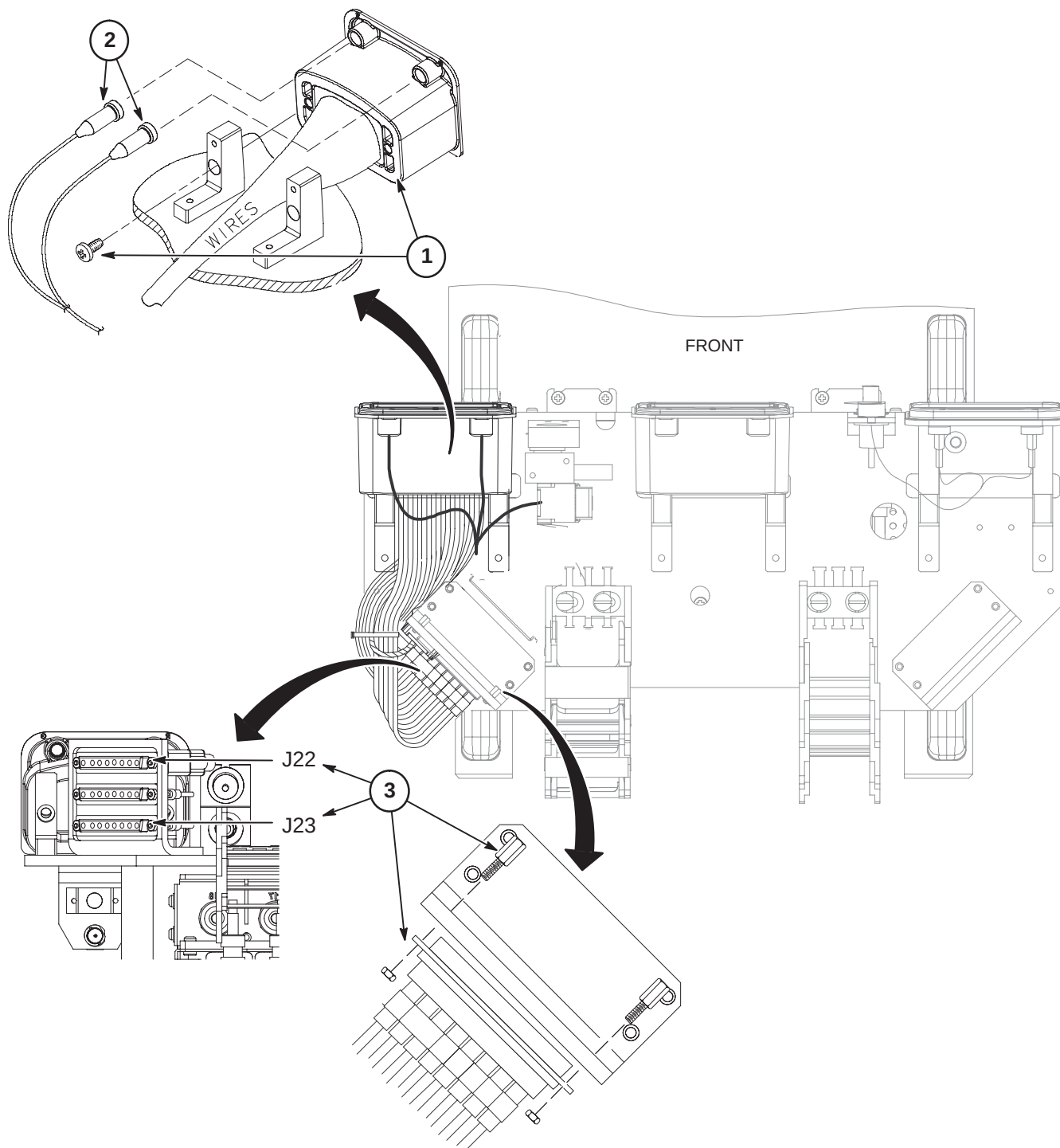
1. Insert the Cable Track assembly into the rear bridge cable track.
2. Remove two screws at location where end of Cable Track attaches to LPCA and apply two drops of LockTite to each screw. Fasten the LPCA end of the cable track to the LPCA with the two screws.
3. Apply two drops of LockTite to the two screws previously removed at the cable track opening in Bridge and attach the other end of the Cable Track.

NOTE: Failure to apply two drops of proper thread locking compound to each cable chain screw before installation could result in screws loosening later and causing spike noise.



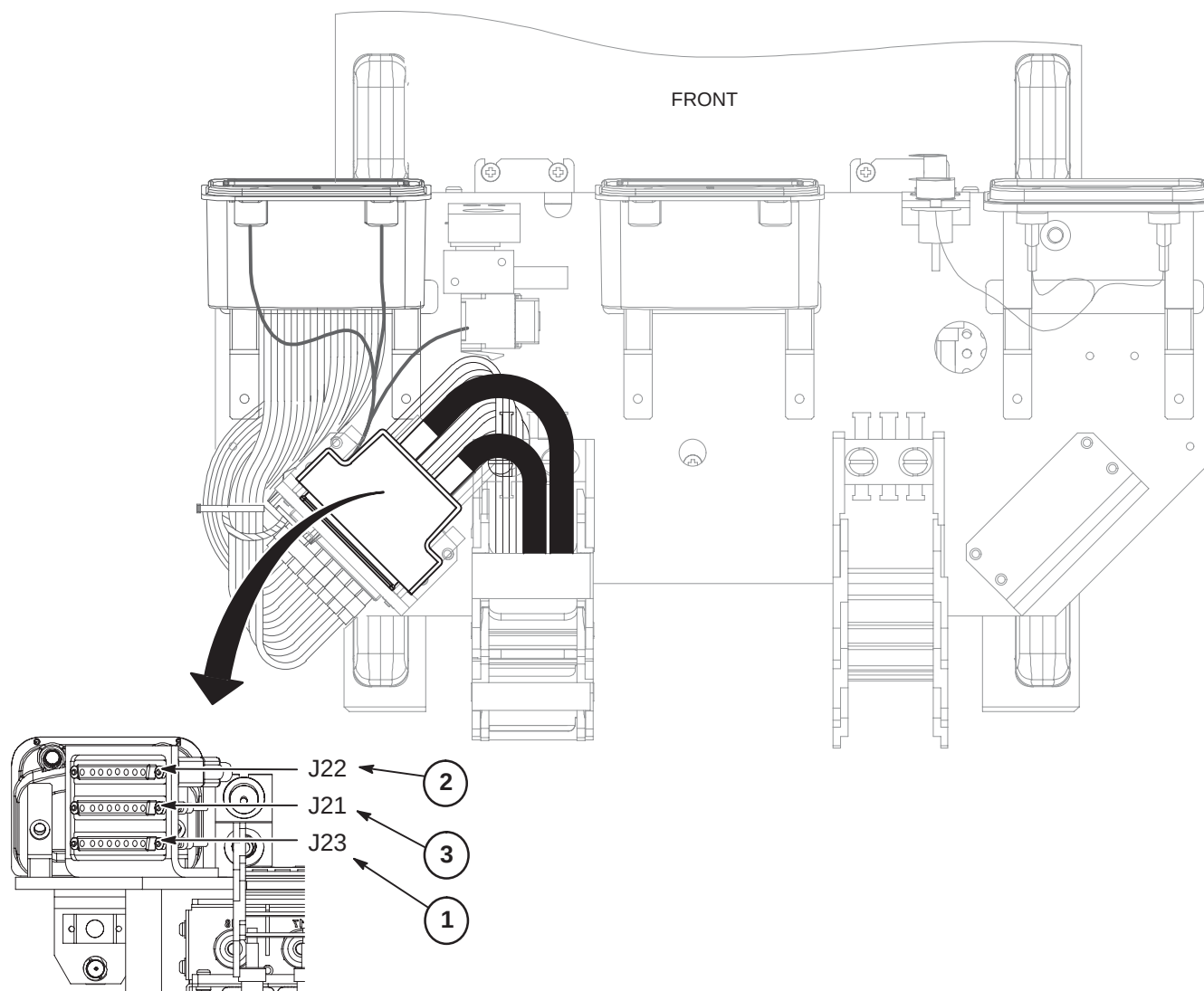
2-5 INSTALL B8-BEZEL IN LPCA

1. Install B8 Bezel and Cable harness (5111020–51) using two M5 x 20mm Brass Screws.
2. Insert the 2 LED's into Bezel.
3. Install 2 sets of D-sub Jackscrews with nuts (5117237) at J22 and J23 as shown and connect harness as marked to J22 and J23 to bracket. There will be an extra set of Jackscrews that are not used for this B8 Bezel.



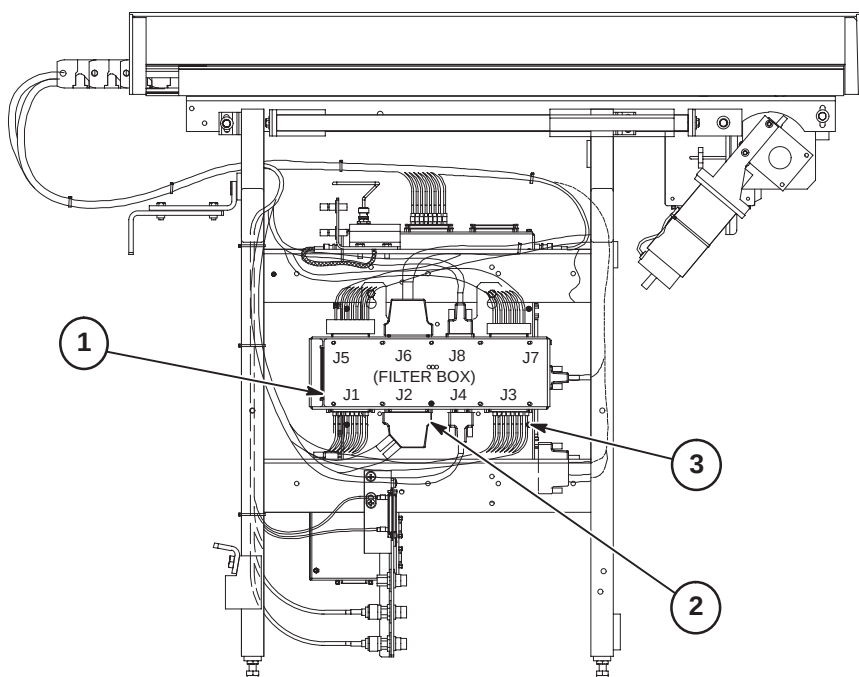
2-6 CONNECT RUNS 1339 AND 1340 TO B8 BEZEL CONNECTOR

1. Connect end of cable run 1339 to J23 on the LPCA bracket.
2. Connect end of cable run 1340 to J22 on the LPCA bracket.
3. Cable run 1341 is not used for this upgrade. The cable can be connected to the blank J21 on the bracket using the extra set of Jackscrews (5117237).



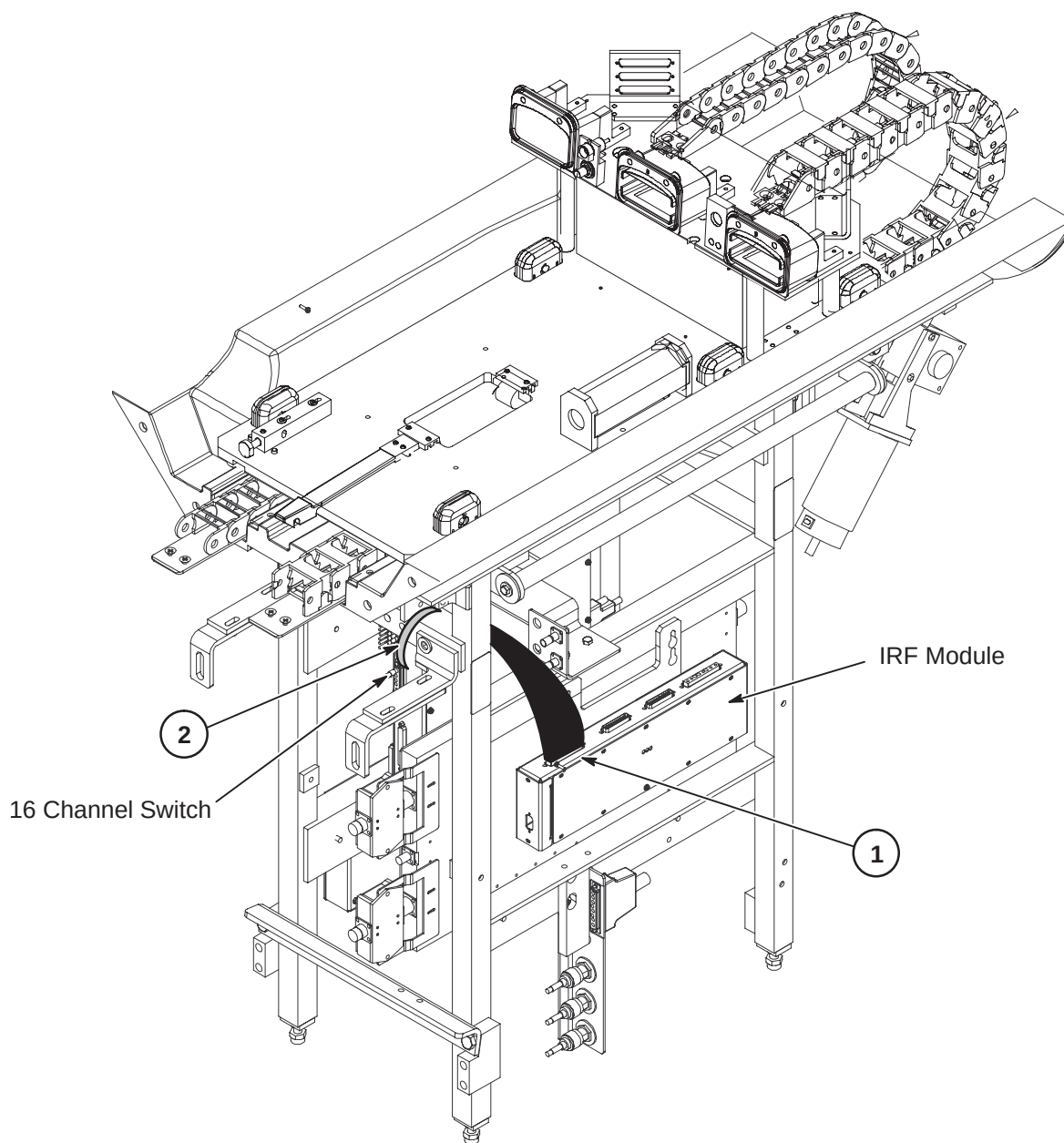
2-7 CONNECT RUNS 1339, 1340, AND 1341 TO IRF MODULE

1. Connect cable run 1339 to J1 on the Image Reject Filter.
2. Connect cable run 1340 to J2 on the Image Reject Filter.
3. Connect cable run 1341 to J3 on the Image Reject Filter.



2-8 INSTALL RUN 1334

1. Connect one end of Run 1334 as marked to J5 on IRF module. Route along upper front bracket of Rear Pedestal and connect to 16 Channel Switch J22.
2. Dress and tie-wrap Run 1334 to upper front bracket of Rear Pedestal.



2-9 INSTALL LPCA COVER

1. Carefully place cover on base plate assembly.
2. Center the front knob within the hole on the x-axis. The y-axis does not need to be centered.
3. Once centered, secure the cover to the base plate using four screws previously removed.
4. Attach Rear Cover using 6 screws previously removed.

